Shora7 Al-Sh3er: Standard RAG vs Graph RAG

An expert system for analyzing and interpreting the Muallaqat Al-Ashr using dual retrieval-augmented generation architectures

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Background & Motivation

Bridging the Gap Between Classical Arabic Poetry and Modern Readers

Over time, many people have grown less connected to classical Arabic, making it increasingly difficult to appreciate the deep meanings and linguistic beauty of the **Muallaqat Al-Ashr** — among the most celebrated poems in Arabic literary heritage.

From this challenge, **Shora7 Al-Sh3er** was born: an intelligent system that makes the Muallaqat accessible, interpretable, and explorable for modern readers.

Dataset Construction

Two OCR technologies were compared — **Mistral AI** and **Qara** — to extract text from PDF books of the Muallaqat Al-Ashr. Mistral AI proved faster and more accurate, and was selected for final processing.

OCR Extraction

Mistral AI processed two PDF books, outperforming Qara in speed and accuracy

Manual Cleaning

Noisy elements removed: headers, page numbers, broken formatting, duplicated lines

Verse Chunking

Each poetic verse treated as an independent chunk to preserve semantic and poetic structure

AL-ASHA | ANTARA IBN SHADDAD | LABID IBN RABIAH

Methodology

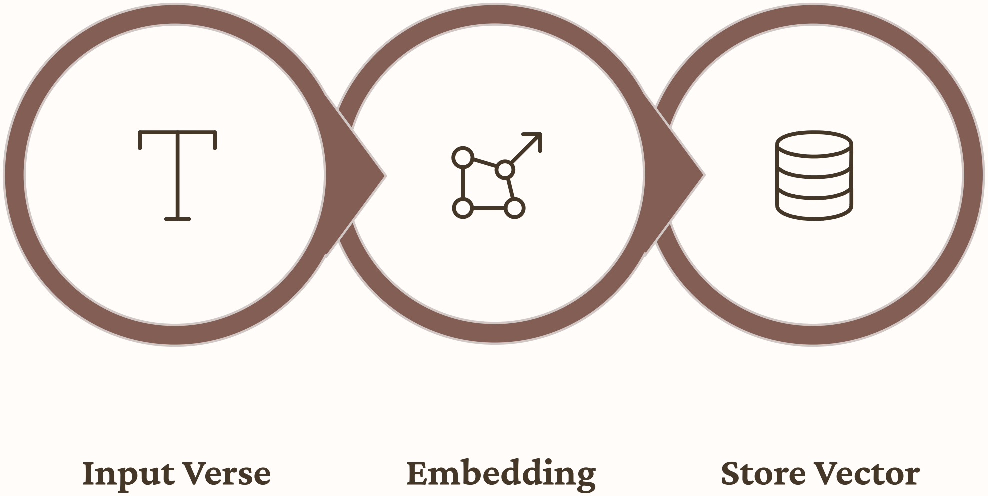
Arabic Embedding Model

Model Used

`Sarah0001/Arabic_embed_model`

A distilled version of `Omartificial-Intelligence-Space/Harrier-Arabic-Matryoshka-270m`, optimized to be lightweight and efficient for retrieval tasks.

> Embeddings help the system understand **semantic meaning and context**, enabling retrieval based on meaning rather than exact keyword matching, which is especially important for classical Arabic poetry.



Architecture 1

Standard RAG Pipeline

01

Query Embedding

User query is converted into a vector using the Arabic embedding model

02

Semantic Search in ChromaDB

System retrieves the most contextually relevant verse chunks via similarity search

03

Answer Generation

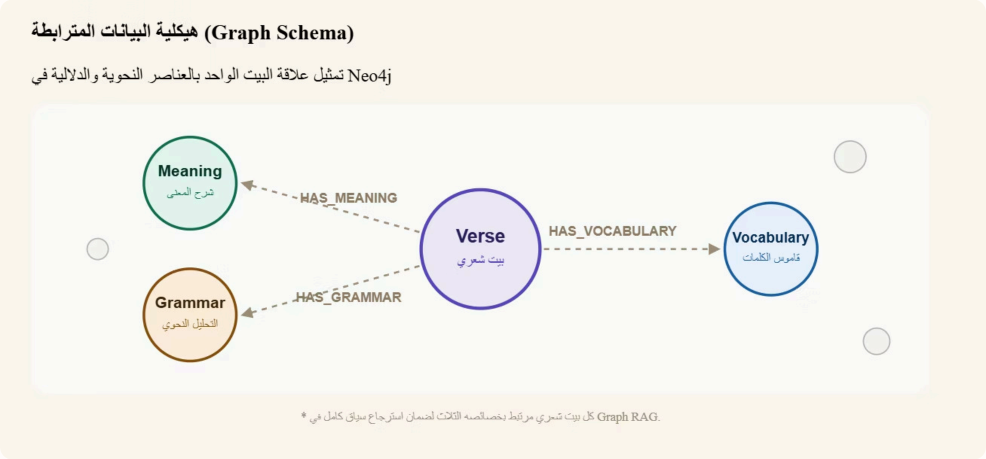
Retrieved context is passed to **GPT-4o-mini**, which generates grounded natural language explanations

Architecture 2

GraphRAG Pipeline

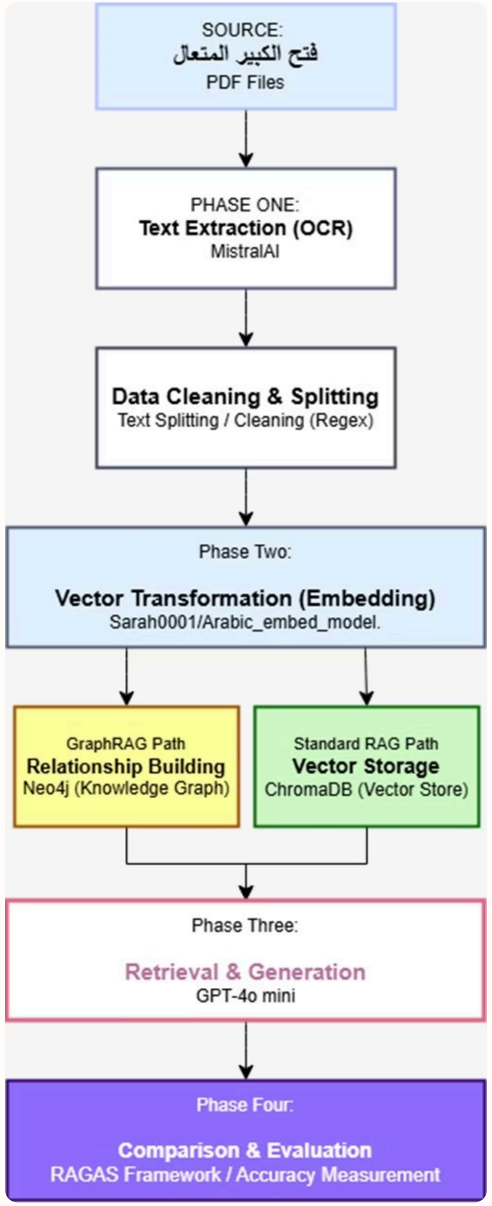
GraphRAG was implemented using **Neo4j** as a knowledge graph database to model structural and semantic relationships within the poetry.

- **GPT-4o-mini** was used to generate grammatical and semantic analyses, which were then stored as interconnected nodes and relationships in Neo4j.
- Graph traversal retrieves **connected, contextually related information** — not isolated chunks
- Final responses are generated using graph-based relational context,



System Design

Dual-Architecture Workflow



Evaluation Framework

RAGAS Metrics & LLM-as-a-Judge

A structured evaluation pipeline was implemented using the **RAGAS framework** alongside a custom **LLM judge** that scores each response from 1–100 across three criteria: accuracy, context adherence, and question relevance.

Faithfulness

Factual consistency with retrieved context — measures hallucination avoidance

Answer Relevancy

How directly the response addresses the user's question

Context Precision

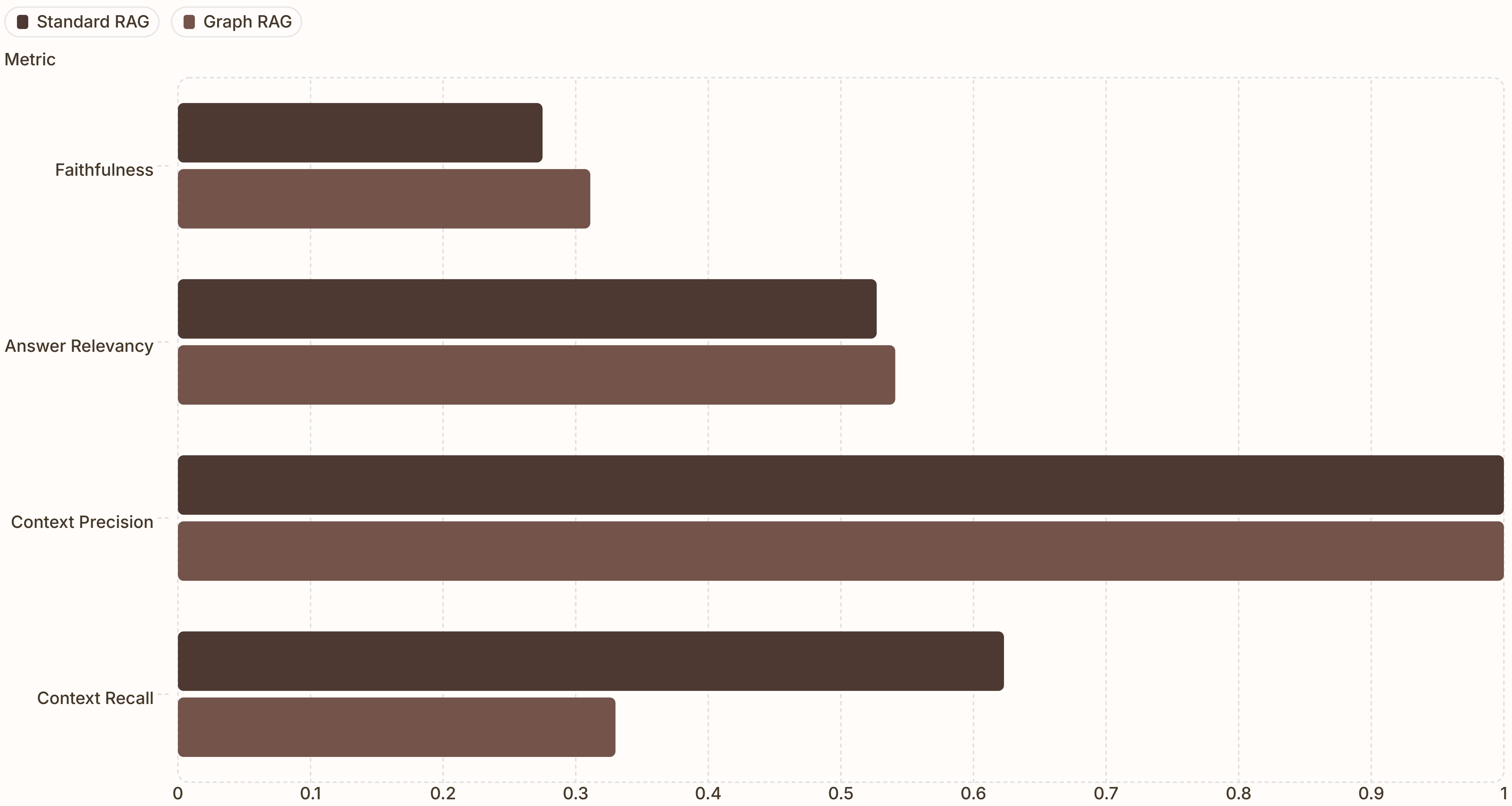
Proportion of retrieved chunks that are genuinely useful

Context Recall

Completeness of retrieved context relative to ground truth

Results

Standard RAG vs GraphRAG — Performance Comparison



Both architectures achieved perfect **Context Precision (1.0)**. GraphRAG now leads in **Faithfulness** and **Answer Relevancy**, while Standard RAG leads in **Context Recall**.

Evaluation Framework

LLM as a Judge Results

A custom LLM judge evaluated both architectures across three criteria: Accuracy (الدقة), Context Adherence (الالتزام بالسياق), and Question Relevance (مدى الارتباط بالسؤال).

Metric	Standard RAG	Graph RAG
Accuracy (الدقة)	75	80
Degree of Adherence (الالتزام بالسياق)	64	70
Relevance (مدى الارتباط)	76	77

Key Finding: GraphRAG demonstrates superior performance in context understanding and question relevance, particularly excelling in complex linguistic analysis tasks.